PRACTICAL - 12

Part A — Practice Programs (Hour 1)

PROBLEM1	1. Word frequency: split a paragraph into words, count each in a HashMap, and print the words sorted by frequency (highest first) using a Comparator.
CODE	

```
import java.util.*;

public class WordFrequency {
   public static void main(String[] args) {

      String paragraph = "java is easy and java is powerful and java is popular";

      HashMap<String, Integer> map = new HashMap<>();

      String[] words = paragraph.toLowerCase().split("\\s+");

      // Count frequency
      for (String word : words) {
         map.put(word, map.getOrDefault(word, 0) + 1);
      }

      // Sort by frequency (highest first)
      List<Map.Entry<String, Integer>> list =
            new ArrayList<>(map.entrySet());

      list.sort((a, b) -> b.getValue().compareTo(a.getValue()));

      // Print result
      System.out.println("Words sorted by frequency:");

      for (Map.Entry<String, Integer> entry : list) {
         System.out.println(entry.getKey() + " : " + entry.getValue());
      }
   }
}
```

OUTPUT

```
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP> cd "Practical 12"
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 12> javac .\WordFrequency.java
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 12> java WordFrequency
Words sorted by frequency:
java : 3
is : 3
and : 2
powerful : 1
easy : 1
popular : 1
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 12>
```

PROBLEM2

2. Unique visitors: from a list of login names with repeats, use a HashSet to count unique users and a LinkedHashSet to keep first-seen order; print the unique count and the first three distinct names.

CODE

```
import java.util.*;

public class UniqueVisitors {
   public static void main(String[] args) {

      List<String> logins = Arrays.asList(
            "Om", "Rahul", "Amit", "Om",
            "Priya", "Rahul", "Neha", "Amit"
      );

      // HashSet for unique count
      HashSet<String> uniqueUsers = new HashSet<>(logins);

      // LinkedHashSet for first-seen order
      LinkedHashSet<String> orderedUsers = new LinkedHashSet<>(logins);

      System.out.println("Unique users: " + uniqueUsers.size());

      System.out.println("First three distinct users:");

      int count = 0;

      for (String user : orderedUsers) {
         System.out.println(user);
         count++;

         if (count == 3)
            break;
      }
   }
}
```

OUTPUT

```
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 12> javac .\UniqueVisitors.java
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 12> java .\UniqueVisitors.java
Unique users: 5
First three distinct users:
Om
Rahul
Amit
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 12>
```

PROBLEM3

3. (For Advanced Learners – home) Build a leaderboard from player scores held in a Map and print the top-N sorted by score (ties broken by name); choose the data and sorting details.

CODE

```
import java.util.*;

public class Leaderboard {
   public static void main(String[] args) {

      Map<String, Integer> scores = new HashMap<>();

      scores.put("Om", 950);
      scores.put("Rahul", 870);
      scores.put("Nishant", 950);
      scores.put("Kushagra", 920);
      scores.put("Neha", 870);

      // Convert Map entries to List
      List<Map.Entry<String, Integer>> players =
            new ArrayList<>(scores.entrySet());

      // Sort by score descending
      // If scores are equal, sort by name
      players.sort((a, b) -> {
         int scoreCompare = b.getValue().compareTo(a.getValue());

         if (scoreCompare != 0)
            return scoreCompare;

         return a.getKey().compareTo(b.getKey());
      });

      int N = 3;

      System.out.println("Top " + N + " Players:");

      for (int i = 0; i < N && i < players.size(); i++) {
         Map.Entry<String, Integer> player = players.get(i);

         System.out.println(
```

```
                (i + 1) + ". " +
                player.getKey() + " - " +
                player.getValue()
            );
        }
    }
}
```

OUTPUT

```
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 12> javac .\Leaderboard.java
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 12> java .\Leaderboard.java
Top 3 Players:
1. Nishant - 950
2. Om - 950
3. Kushagra - 920
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 12>
```